EduGenie: Google Gemini Powered Learning Assistant

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1. Project Overview

EduGenie is a learning assistant concept that uses Google Gemini to help students understand lessons, ask questions, receive simplified explanations, create practice questions, and support personalized learning.

2. Hardware Prerequisites

- Computer or laptop with a modern web browser.
- Smartphone can be used for testing and demonstration.
- Stable internet connection.
- Adequate storage for project files and documentation.

3. Software Prerequisites

- Operating system such as Windows, Linux, or macOS.
- Modern browser such as Chrome, Edge, or Firefox.
- A suitable code editor or IDE.
- Python or JavaScript/TypeScript, depending on the selected implementation.
- Google GenAI SDK when integrating the Gemini API programmatically.

4. Knowledge Prerequisites

- Basic computer and internet knowledge.
- Basic programming concepts such as variables, functions, conditions, and loops.
- Basic knowledge of HTML, CSS, and JavaScript if a web interface is developed.
- Basic understanding of databases if student progress or questions are stored.
- Basic understanding of Artificial Intelligence, Generative AI, and prompt engineering.

5. Gemini / AI Prerequisites

- A Google Account for accessing Gemini services where required.
- Understanding of how prompts are written and refined.
- Awareness that AI-generated responses should be checked for accuracy.
- Responsible use of AI, including privacy and appropriate handling of student information.

6. Project Development Requirements

- Define the target users and learning problems to be addressed.
- Design the EduGenie user interface and learning workflow.
- Connect the application to Gemini using an appropriate Google AI integration.

- Implement features such as question answering, topic explanation, summarization, and practice-question generation.
- Test responses with different subjects and prompts before demonstration.

7. Expected Outcome

The completed EduGenie project should provide a simple AI-assisted learning environment in which students can interact with learning content, ask questions, receive explanations, and practise concepts with Gemini-powered assistance.

Reference Note

Google states that Gemini for Education can support personalized learning, explanations, practice tests, quizzes, and other educational activities. Google also provides an official Google GenAI SDK for programmatic Gemini API integration. These references can be used to justify the project prerequisites and technology choice.